

PWM IP Core

Avalon-MM Peripheral — Cyclone V FPGA

Revision 1.0 | Duval MAMBOU | ENIB — Bretagne INP

Interface Avalon-MM Slave	Target device Cyclone V (5CSEMA4U23C6)	Clock 25 MHz
Data width 32 bits	Address width 3 bits	Registers 6 (5 addresses)

Overview

The PWM IP Core is a 32-bit Avalon Memory-Mapped slave peripheral that generates a Pulse Width Modulation (PWM) signal with programmable period, duty cycle, and interrupt support. It is designed to be integrated into Intel/Altera Qsys SoC systems alongside a Nios II processor.

The IP implements a synchronous architecture with a two-stage reset synchronizer and supports software control via a C API. The PWM output signal frequency and duty cycle are fully configurable through memory-mapped registers accessible over the Avalon-MM bus.

Key features:

- Programmable PWM period and duty cycle (32-bit resolution)
- Shadow register mechanism for glitch-free synchronous parameter updates
- Configurable interrupt generation after N PWM cycles
- Read-on-clear interrupt request flag
- Two-stage synchronous reset
- Avalon-MM compliant slave interface (no wait states)

Port Description

Port	Width	Direction	Description
Clk	1	Input	System clock
Reset_n	1	Input	Active-low asynchronous reset
WritePWM	1	Input	Avalon write strobe
ReadPWM	1	Input	Avalon read strobe
RegsAddr	3	Input	Register address (offsets 0–4)
DInPWM	32	Input	Avalon write data bus
DOutPWM	32	Output	Avalon read data bus
OutPWM	1	Output	PWM output signal
IrqPWM	1	Output	Interrupt request output

Register Map

Offset	Name	Access	Reset value	Bits used	Description
0x0	Status	RO	0x00000000	[2:0]	PWM state, pulse state, IRQ request
0x0	SetCtrl	WO	—	[1:0]	Set Start and IrqEn bits
0x1	Period	R/W	0xFFFFFFFF	[31:0]	PWM period value (clock cycles)
0x2	DCycle	R/W	0x00000000	[31:0]	Duty cycle value (clock cycles)
0x3	NbCycles	R/W	0x0000FFFF	[15:0]	Number of PWM cycles before IRQ
0x4	ClrCtrl	WO	—	[1:0]	Clear Start and IrqEn bits

Note: Status and SetCtrl share address offset 0x0. A read access targets Status; a write access targets SetCtrl.

Status Register (offset 0x0 — Read Only)

Bit	Field	Description
0	PWMState	Reflects the current value of the Start bit. Set to 1 when the PWM is running.
1	PulseState	Reflects the current state of the PWM output signal (high or low).
2	IrqRq	Interrupt request flag. Set when CntCyc reaches NbCyc. Cleared automatically on read.
31:3	—	Reserved, read as 0.

SetCtrl / ClrCtrl Registers (offsets 0x0 / 0x4 — Write Only)

Bit	Field	Description
0	Start	Writing 1 sets (SetCtrl) or clears (ClrCtrl) the Start bit of the internal Control register.
1	IrqEn	Writing 1 sets (SetCtrl) or clears (ClrCtrl) the IrqEn bit of the internal Control register.

The internal Control register is not directly accessible. It is modified exclusively through SetCtrl and ClrCtrl using a set/clear mechanism: writing a 1 to a bit in SetCtrl forces that bit to 1 in Control; writing a 1 to the same bit in ClrCtrl forces it to 0.

Period Register (offset 0x1 — Read/Write)

Defines the PWM period in clock cycles. The actual PWM period in seconds is:

$$T_{PWM} = \frac{\text{Period} + 1}{f_{Clk}}$$

Writing a new value to Period automatically resets DCycle to 0 to prevent inconsistent duty cycle configurations.

DCycle Register (offset 0x2 — Read/Write)

Defines the duty cycle in clock cycles. The PWM output is high for DCycle clock cycles out of each period. The duty cycle ratio is:

$$\alpha = \frac{\text{DCycle}}{\text{Period}}$$

For a 25% duty cycle with Period = 1000: set DCycle = 250.

NbCycles Register (offset 0x3 — Read/Write)

Specifies the number of complete PWM periods that must elapse before asserting the interrupt request. Only bits [15:0] are used; bits [31:16] are ignored on write and read as 0.

Functional Description

Block Diagram

Figure 1 shows the complete hardware architecture of the PWM IP core, including the datapath, control logic, decoder, and register bank.

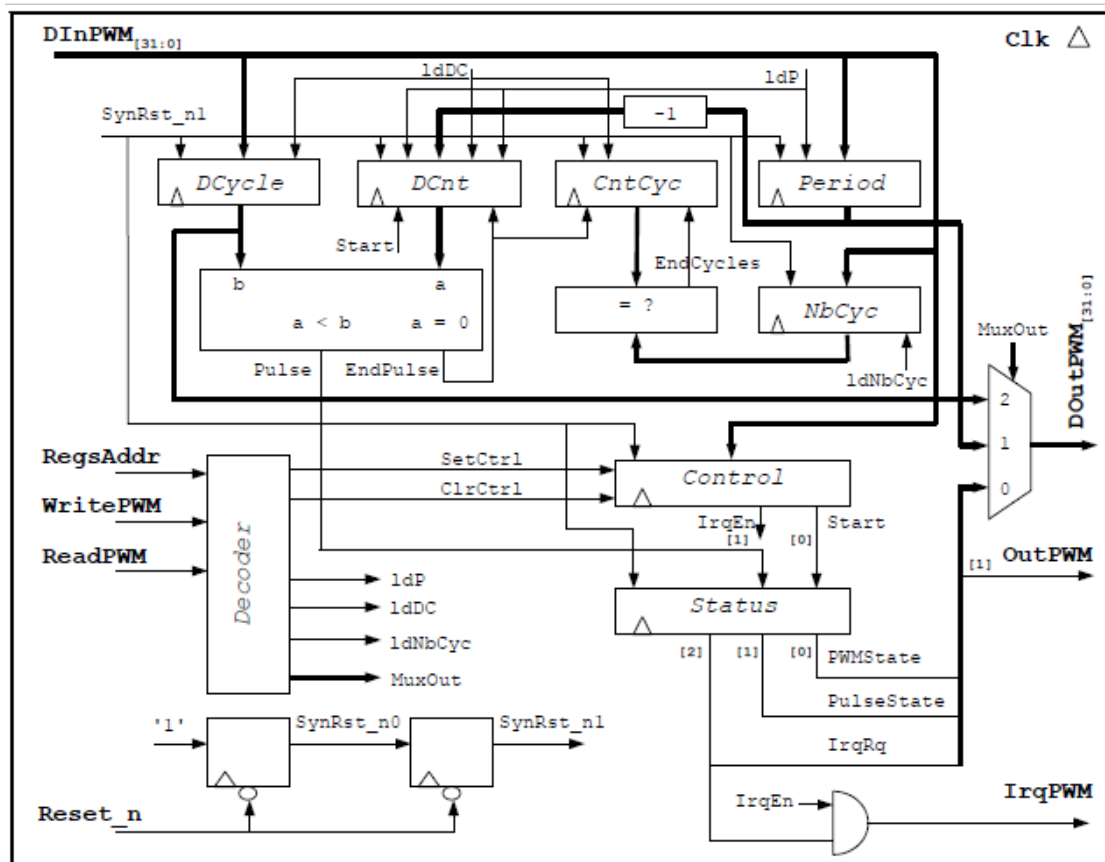


Figure 1: PWM IP Core — Hardware Architecture

Reset Architecture

The IP implements a two-stage synchronous reset synchronizer. The active-low asynchronous **Reset_n** input is re-synchronized to the clock domain through two cascaded flip-flops (**SynRst_n0** and **SynRst_n1**). All internal registers are reset on **SynRst_n1** = '0', except **DCnt** which is synchronously reset only.

PWM Generation

The PWM signal is generated using an internal 32-bit down-counter (**DCnt**). Each clock cycle, **DCnt** decrements by 1. When **DCnt** reaches 0, one complete PWM period has elapsed (**EndPulse** = '1') and **DCnt** is reloaded with **Period** - 1.

The output signal **Pulse** is asserted high when **DCnt** < **DCycle**, producing the high phase of the PWM waveform. The registered version of **Pulse** drives the **OutPWM** output port via **PulseState**.

Shadow Register Mechanism

Updates to **Period** and **DCycle** are applied synchronously at period boundaries. Writing a new **Period** value immediately resets **DCnt** and clears **DCycle** to 0, ensuring the new configuration takes effect at the start of the next period without producing a glitch on the output signal.

Cycle Counter and Interrupt

An internal 16-bit cycle counter (**CntCyc**) increments on each **EndPulse** event. When **CntCyc** reaches the value stored in **NbCyc**, the **EndCycles** signal is asserted, which sets the **IrqRq** flag in the Status register and resets **CntCyc** to 0.

The interrupt output **IrqPWM** is asserted only when both **IrqRq** = '1' and **IrqEn** = '1':

$$\text{IrqPWM} \leq \text{IrqRq} \text{ and } \text{IrqEn}$$

The **IrqRq** flag is cleared automatically when the **Status** register is read (**ReadPWM** = '1' and **RegsAddr** = "000").

Address Decoder

The decoder generates internal control signals from the Avalon bus signals **WritePWM**, **ReadPWM**, and **RegsAddr**:

Signal	Condition	Effect
SetCtrl	Write, addr = 000	Set bits in Control register
ldP	Write, addr = 001	Load Period register
ldDC	Write, addr = 010	Load DCycle register
ldNbCyc	Write, addr = 011	Load NbCycles register
ClrCtrl	Write, addr = 100	Clear bits in Control register
razIrq	Read, addr = 000	Clear IrqRq flag

Timing Characteristics

Parameter	Value	Condition
Target clock frequency	25 MHz	
Clock period	40 ns	
Setup slack (worst case)	+36.466 ns	Slow corner
Hold slack (worst case)	+0.135 ns	Fast corner
Timing closure	Achieved	
Read wait states	0	
Write wait states	0	

Tool: Quartus Prime 17.0 — TimeQuest Timing Analyzer

Device: Cyclone V 5CSEMA4U23C6 — DE0-CV development board

SoC Integration

The IP is packaged as a Qsys component with the following interfaces:

- **avalon_slave_0** — Avalon-MM Slave (32-bit data, 3-bit address, 0 wait states)
- **clock** — Clock input interface
- **reset** — Reset input interface (active-low, asynchronous)
- **interrupt_sender** — Interrupt sender interface (associated to `avalon_slave_0`)
- **conduit_end** — Conduit export for OutPWM signal

The OutPWM signal is exported from the Qsys system under the name `pwm_out_sig_export` and can be connected to external hardware (e.g., 7-segment display enable signals for brightness control).

The IP was validated in a complete Qsys SoC system including a Nios II processor, operating across a Clock Domain Crossing Bridge at 25 MHz.

Simulation

Functional simulation was performed under ModelSim using TCL macro scripts (`.do` files) emulating Avalon-MM transactions. Four test scenarios were executed:

1. **Register access verification** — read/write access to all registers
2. **PWM generation** — duty cycles of 25% and 75%
3. **Interrupt behavior** — IRQ generation every 10 cycles, and IRQ disarm
4. **Boundary conditions** — duty cycles of 0%, 1%, 99%, and 100%

Post-synthesis structural simulation was performed on the placed-and-routed netlist to confirm functional equivalence with the RTL simulation results.

Revision History

Rev.	Date	Author	Description
1.0	April 2026	Duval MAMBOU	Initial release